A Crossover Trial of Hospital-Wide Lactated Ringer's Solution versus Normal Saline

Critical Appraisal of the FLUID Trial

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RCT

ORIGINAL ARTICLE

A Crossover Trial of Hospital-Wide Lactated Ringer's Solution versus Normal Saline

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Trial Identity and Citation

TRIAL NAME & CITATION

FLUID Trial: A Crossover Trial of Hospital-Wide Lactated Ringer's Solution versus Normal Saline
Citation: N Engl J Med 2025;393:660-70

FUNDING & RESEARCH GROUPS

- Full Citation: McIntyre L, Fergusson D, McArdle T, et al. A Crossover Trial of Hospital-Wide Lactated Ringer's Solution versus Normal Saline. N Engl J Med 2025;393:660-670. doi:10.1056/NEJMoa2416761
- Trial Name: FLUID trial.
- Trial Registration: ClinicalTrials.gov number, NCT04512950.

TRIAL REGISTRATION & SPONSORS

- Funding: Funded by the Canadian Institutes of Health Research and the Ottawa Hospital Academic Medical Organization.
- Sponsors: Investigator-initiated, academic study coordinated by the Ottawa Methods Centre. Baxter Healthcare provided study fluids but had no role in trial design, conduct, analysis, or writing.

Structure of This Appraisal

1. **Introduction**
   Clinical question, background, and rationale for crystalloid choices.

2. **Methods**
   Pragmatic cluster-randomized crossover design, eligibility, and statistical plan.

3. **Results**
   Participant baseline characteristics, primary composite outcome, secondary outcomes, and subgroup findings.

4. **Discussion**
   Methodological strengths, critical limitations, post-hoc Bayesian reanalysis, and local applicability to North Indian / Ghaziabad patients.

INTRODUCTION

Background, Clinical Need, and Trial Rationale

Why This Paper Matters

Clinical problem: Crystalloid fluids are among the most common interventions in hospitalized patients, but normal saline has high chloride content which can cause hyperchloremic metabolic acidosis.

Evidence gap: Large trials show balanced crystalloids improve major adverse kidney events in critical care, but whether hospital-wide use reduces patient-important clinical outcomes outside the ICU is uncertain.

Research question: Does a hospital-wide policy of using lactated Ringer's solution compared to normal saline reduce death or readmission within 90 days after index admission?

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Clinical Question & PICO Framework

P — Hospitalized patients in academic and community hospitals requiring intravenous fluids, admitting >=1500 patients per period, with level II/III ICU capacity.

I — Hospital-wide policy of administering Lactated Ringer's solution from the time of admission through discharge.

C — Hospital-wide policy of administering Normal Saline (0.9% NaCl) as the main crystalloid fluid.

O — Composite of death from any cause or readmission to the hospital within 90 days after index admission.

METHODS

Study Population, Eligibility, Outcomes, and Statistics

Study Design & Methodology

TRIAL ARCHITECTURE

- Design: Investigator-initiated, open-label, pragmatic, two-period, two-sequence, cross-sectional, cluster-randomized, crossover trial.
- Crossover Sequence: Hospitals assigned to either LR followed by saline, or saline followed by LR. Each active period lasted 12 weeks, with a 1-2 week washout period.
- Settings: 7 academic and community hospitals in Ontario, Canada, representing a diverse patient population.

BIAS CONTROLS

- Consent & Blinding: Waiver of individual patient consent due to minimal risk. Open-label for clinicians/patients, but outcomes gathered from population administrative databases (ICES) and analyzed independently.
- Adherence Monitoring: Tracked hospital logistic service inventory daily. automatic substitution orders in EHR or manual order entry forced the assigned fluid.

Pragmatic cluster crossover: Randomization occurred at the hospital level. An automatic substitution order was executed hospital-wide.

Eligibility Criteria

INCLUSION CRITERIA

- Hospitals: Admitting >=1500 patients per trial period and having level II or III ICU capacity.
- Patients: All patients with an index admission to the trial hospital during the active weeks of the study period.
- Fluid Exposure: Patients eligible regardless of actual exposure to the trial fluid (pragmatic, ITT approach).

EXCLUSION CRITERIA

- Age Constraint: Patients younger than 1 month of age were excluded.
- Data Quality: Excluded patients with missing birth dates or those not meeting index admission definition.
- Period Constraints: Admissions during the run-in or washout periods were excluded from the analysis.

Index Admission: Patient's first admission with no prior hospital admission in the previous 90 days. Avoided multiple counting of the same patient.

Trial Endpoints & Definitions

PRIMARY OUTCOME

- A composite of death from any cause or readmission to the hospital within 90 days after index admission.
- Data obtained and validated with >99% accuracy in the ICES administrative databases.

SECONDARY OUTCOMES

- Individual components: death within 90 days; hospital readmission within 90 days.
- Clinical outcomes: initiation of dialysis within 90 days; emergency department visit within 90 days.
- Health system outcomes: hospital length of stay (hours); discharge to a facility other than home.

SAFETY OUTCOMES

- Serious adverse events (SAEs) related to fluid administration.
- Reported via site principal investigators and routinely scheduled safety committee meetings or morbidity & mortality rounds.

Statistical Plan & Power

16 sites
original target sample size

7 sites
actual randomized sites

80%
power to detect 1% absolute diff

ICC 0.0064
within-period correlation

- Sample Size target: 16 hospitals would yield ~144,000 admissions. Provide 80% power to detect a 1% absolute difference (16% standard vs 15% intensive/LR).
- Early Termination: Stopped after 7 hospitals completed both periods because of infeasibility during the Covid-19 pandemic.
- Primary Analysis: Model trial period as binary; linear regression with period, intervention, and hospital as independent variables. Degrees of freedom = clusters - 2 (5 d.f.).
- Adjustments: Two-stage Hayes-Moulton method adjusting for age, sex, case-mix group, Elixhauser comorbidity score, and ICU admission within 1 day.

Protocol Adherence and Achieved Exposure

78.2%
LR adherence

93.6%
Normal Saline adherence

43,626
total patients analyzed

0
missing primary outcome data

- Asymmetric Adherence: Adherence was lower in the Lactated Ringer's group (78.2%) compared to the Normal Saline group (93.6%).
- Fluid contamination: Non-assigned fluids accounted for 21.8% of use in the LR periods, representing considerable treatment contamination.
- Data Completeness: High data integrity with zero missing values for the primary composite outcome or secondary outcomes.
- Analysis Set: All eligible patients analyzed under the ITT principle, regardless of actual fluid exposure.

Methods Critical Appraisal: Internal Validity

METHODOLOGICAL STRENGTHS

- Cluster Randomization: Sequentially generated block randomization successfully balanced cluster characteristics.
- Data Source: Use of validated, population-based health registries minimized loss to follow-up and detection bias.
- Blinded Analysis: ICES analysts were blinded to randomized sequences, reducing reporting and analytical bias.

LIMITATIONS & CONCERNS

- Early Stopping: Stopped at 7 of 16 planned clusters, leaving the study underpowered to detect a 0.5% to 1.0% absolute risk reduction.
- Asymmetric Adherence: LR group had 21.8% non-compliance, diluting the treatment effect toward the null.
- Open-Label Design: Open-label policy could theoretically introduce co-intervention bias, though hospital-wide policies minimized this.

Overall Methods Appraisal: A pragmatic, well-designed trial with a waiver of consent, but significantly limited by early termination and asymmetric fluid adherence.

RESULTS

Enrollment, Baseline Characteristics, and Outcome Data

Figure 1. Trial Design Schema

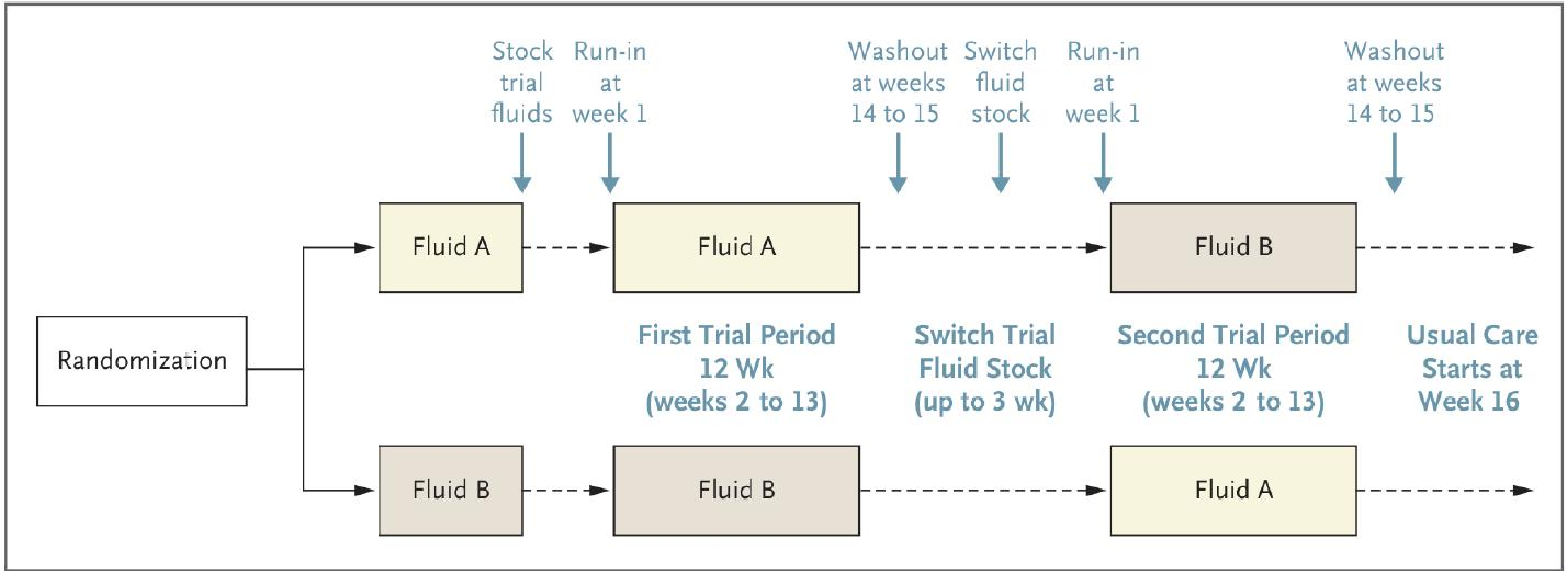


Figure 1. Trial Design.

The trial included two active 12-week periods. Before each period, we allotted time (1 to 3 weeks) for participating sites to stock the assigned trial fluid throughout the hospital, after which there was a 1-week run-in period to give hospital staff (physicians and nurses) time to familiarize themselves with trial operations and the automatic substitution order; no patients were included in the analysis during the 1-week run-in period. The run-in period was followed by a 12-week active trial period (weeks 2 through 13); all patients with an index hospital admission during this time were included in the analysis unless they met criteria for exclusion. Each 12-week active trial period was followed by a washout period lasting either 1 week (sites 1 through 5) or 2 weeks (sites 6 and 7) to ensure that patients who were admitted close to the end of the active trial period received the same trial fluid throughout their hospital stay; no patients admitted during the washout periods were included in the analysis.

Figure 1. Crossover design with two 12-week active periods, run-in weeks, and washout periods.

Figure 2. Participant Flowchart

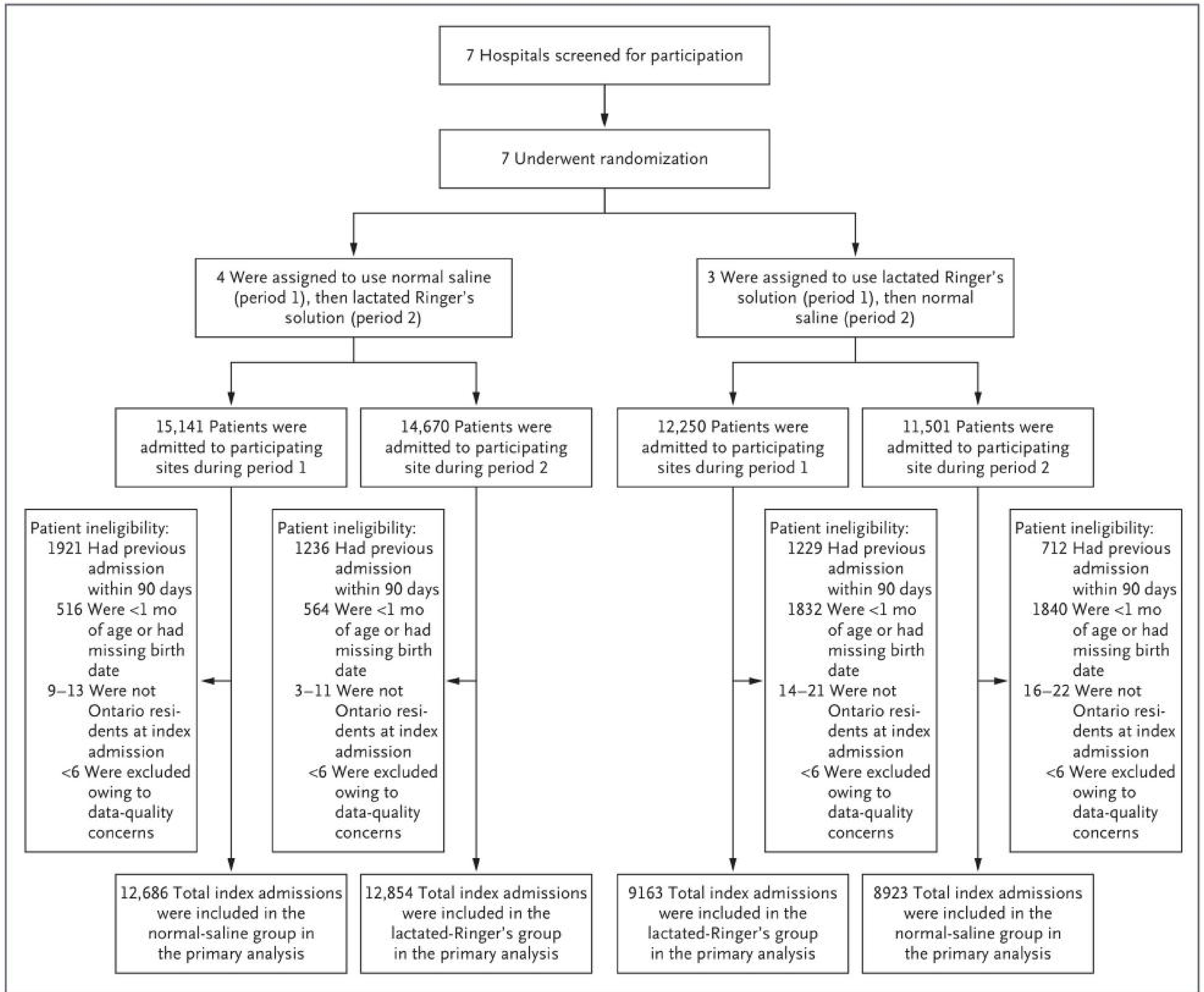


Figure 2. **Screening, Randomization, and Analysis.**

Where ranges are given, exact values are suppressed owing to data privacy or a data sharing agreement between ICES and data providers. In the case of patients excluded owing to data-quality concerns, the validity of the date of death is unclear because the database indicates that the date of death occurred before the index admission. These patients were excluded from analyses according to standard ICES procedures.

Figure 2. Trial screening and retention flow: 43,626 patients analyzed across 7 hospitals.

Table 1. Characteristics of the Patients at Baseline

Table 1. Characteristics of the Patients at Baseline.*

Characteristic	Lactated Ringer's Solution (N=22,017)	Normal Saline (N=21,609)
Age — yr	58.5±20.6	58.7±20.8
Female sex — no. (%)	12,682 (57.6)	12,623 (58.4)
Place of residence — no (%)†		
Rural	2,221 (10.1)	2,092 (9.7)
Urban	19,743 (89.7)	19,471 (90.1)
Unknown	53 (0.2)	46 (0.2)
Income quintile — no. (%)‡		
1	4,874 (22.1)	4,828 (22.3)
2	4,549 (20.7)	4,574 (21.2)
3	4,362 (19.8)	4,267 (19.7)
4	4,066 (18.5)	4,094 (18.9)
5	4,087 (18.6)	3,778 (17.5)
Unknown	79 (0.4)	68 (0.3)
Case-mix group — no. (%)§		
Medicine	8,787 (39.9)	9,011 (41.7)
Surgery	9,379 (42.6)	8,804 (40.7)
Pregnancy and childbirth	3,706 (16.8)	3,670 (17.0)
Mental health	145 (0.7)	124 (0.6)
Elixhauser Comorbidity Index score¶		
Mean	2.6±5.2	2.8±5.3
Median (IQR)	0 (0–5)	0 (0–5)
Elixhauser Comorbidity Index quartile‖		
1	1,399 (6.4)	1,185 (5.5)
2	12,671 (57.6)	12,482 (57.8)
3	2,219 (10.1)	2,090 (9.7)
4	5,728 (26.0)	5,852 (27.1)

* Plus–minus values are means ±SD. Percentages may not total 100 because of rounding. IQR denotes interquartile range.

† Place of residence was determined by Statistics Canada with the use of census metropolitan area (CMA) or census agglomeration (CA) data.

‡ Income quintiles were determined on the basis of income data for the nearest Canadian census-based neighborhood within a CMA or CA. Lower numbers indicate lower incomes.

§ Case-mix groups were defined according to Canadian Institute for Health Information methods that use similarities in diagnoses, interventions, length of stay, and resource use to stratify patients receiving acute care in an inpatient setting.

¶ The Elixhauser Comorbidity Index is a composite score that weights individual coexisting conditions (referred to as comorbidities) according to their independent associations with mortality. The composite score can be negative or positive because some of the conditions are inversely associated with mortality. Scores range from −19 to 89, with higher scores indicating a greater burden of diseases associated with an increased risk of in-hospital death.

‖ Higher numbers indicate higher scores on the Elixhauser Comorbidity Index.

P=0.35) and a cluster-level adjusted relative risk of 0.97 (95% CI, 0.90 to 1.05; P=0.35) (Table 2). dar time) showed results consistent with those of the primary analysis (Table S2).

Table 1. Baseline demographic and clinical characteristics: age, sex, residence, income, and comorbidities.

Baseline Comparability Critique

- Patient Characteristics: Mean age was 58.5 years in the LR group and 58.7 years in the normal saline group. Female sex was 57.6% (LR) vs 58.4% (saline).
- Socioeconomic Balance: Income quintile distributions were well matched. Case-mix groups (Medicine ~40%, Surgery ~41%, Pregnancy/childbirth ~17%) were identical.
- Comorbidity Index: Mean Elixhauser Comorbidity Index was balanced: 2.6 in the LR group vs 2.8 in the saline group.
- Imbalance Analysis: No clinically or statistically meaningful differences in baseline characteristics were present, confirming the success of cluster randomization.

Verdict: Randomization succeeded. Baseline characteristics: mean age (~58.6 years), female sex (~58%), rural residence (~10%), and comorbidities were exceptionally balanced.

Table 2. Summary of Outcomes

Table 2. **Summary of Cluster-Level Analyses of Primary and Secondary Outcomes.***

Outcome	Lactated Ringer's Solution (N=7)	Normal Saline (N=7)	Adjusted Mean Difference (95% CI)†	Adjusted Relative Difference (95% CI)
Primary outcome				
Composite of death or readmission within 90 days after index admission (%)	20.3±3.5	21.4±3.3	−0.53 (−1.85 to 0.79)‡	0.97 (0.90 to 1.05)
Secondary outcomes				
Death within 90 days after index admission (%)	6.9±1.2	7.6±1.7	−0.26 (−0.95 to 0.43)	0.97 (0.89 to 1.07)
Readmission to hospital within 90 days after index admission (%)	15.1±2.9	15.4±2.1	−0.06 (−1.78 to 1.67)	0.99 (0.87 to 1.11)
Emergency department visit within 90 days after index admission (%)§	21.2±3.1	21.0±2.3	0.28 (−2.51 to 3.06)	1.01 (0.88 to 1.15)
Initiation of dialysis within 90 days after index admission (%)	0.5±0.3	0.6±0.5	−0.12 (−0.34 to 0.11)	0.99 (0.56 to 1.77)
Hospital length of stay (hr)	164.7±34.7	172.3±34.2	−0.002 (−0.006 to 0.002)	NA¶
Discharge to a facility other than home (%)	15.4±4.8	16.2±4.6	−0.49 (−1.59 to 0.62)	0.96 (0.90 to 1.03)

* Percentages are the incidence of primary- and secondary-outcome events averaged across seven hospitals (clusters). Hospital length of stay is also averaged across hospitals. Plus–minus values are means ±SD. CI denotes confidence interval.

† Values are from linear regression analysis of cluster-level summary data adjusted for age, sex, case-mix group, Elixhauser Comorbidity Index score, and admission to the intensive care unit within 1 day after hospital admission. All analyses account for a binary effect of trial period (first or second) and clustering according to hospital. Values are expressed as percentage points, except in the case of hospital length of stay, in which the adjusted mean difference is expressed in hours.

‡ P=0.35.

§ Numbers include only the first visit to an emergency department within 90 days after the index admission.

¶ The adjusted relative difference is not applicable (NA) for hospital length of stay because length of stay is a continuous outcome.

Table 2. Efficacy results: composite of death or readmission within 90 days, individual components, and other secondary outcomes.

Efficacy Results Interpretation

20.3% vs 21.4%
primary outcome (LR vs Saline)

-0.53%
adjusted mean diff (95% CI: -1.85% to 0.79%)

P = 0.35
statistically non-significant

0.97
relative risk (95% CI: 0.90 to 1.05)

- Primary Composite Outcome: Composite of death or readmission within 90 days occurred in 20.3% (LR) vs 21.4% (Saline). Adjusted difference -0.53 percentage points (p = 0.35).
- Relative Effect: Relative Risk of 0.97 (95% CI, 0.90 to 1.05) indicates no statistically significant difference between groups.
- Effect Direction: The point estimate slightly favored Lactated Ringer's, but the wide confidence interval includes both potential benefit and potential harm.
- Pragmatic Interpretation: A hospital-wide policy of substituting normal saline with lactated Ringer's does not yield a significant decrease in 90-day death or readmission.

Secondary Efficacy and Clinical Outcomes

- Death within 90 days: Occurred in 6.9% of LR patients vs 7.6% of Saline patients (adjusted difference, -0.26 percentage points; 95% CI, -0.95 to 0.43; RR 0.97 [0.89 to 1.07]).
- Readmission within 90 days: 15.1% in the LR group vs 15.4% in the Saline group (adjusted difference, -0.06 percentage points; 95% CI, -1.78 to 1.67; RR 0.99 [0.87 to 1.11]).
- Initiation of Dialysis: 0.5% (LR) vs 0.6% (Saline) (adjusted difference, -0.12 percentage points; 95% CI, -0.34 to 0.11; RR 0.99 [0.56 to 1.77]). Neutral result.
- Emergency Department Visits: 21.2% in LR vs 21.0% in Saline. Adjusted difference, 0.28 percentage points (95% CI, -2.51 to 3.06; RR 1.01 [0.88 to 1.15]).
- Health System Outcomes: Hospital length of stay (164.7 vs 172.3 hours) and discharge to facility other than home (15.4% vs 16.2%) were similar.

Safety and Adverse Events

- Reporting System: Sites monitored SAEs through routinely scheduled safety committee meetings and morbidity/mortality rounds.
- SAE Count: Zero serious adverse events were reported to the central coordinating site in both arms.
- Lack of Lab Harms Data: Note that the trial did not report specific lab-based harms, such as incidence of hyperkalemia or metabolic acidosis, representing a critical data gap.

Safety Verdict: No serious adverse events related to fluid administration were reported by any participating hospitals, indicating excellent tolerability of both fluids.

Figure 3. Subgroup Analyses

Subgroup	No. of Patients	Lactated Ringer's Solution	Normal Saline	Risk Difference (95% CI)
		estimated incidence		*percentage points*
Sex				
Female	25,305	18.2±3.9	19.4±4.6	−1.11 (−2.49 to 0.28)
Male	18,321	23.7±3.2	25.0±3.2	−1.25 (−2.98 to 0.48)
Age				
1 mo to 18 yr	198	8.7±4.8	10.4±10.3	−1.74 (−12.4 to 8.92)
>18 to 65 yr	24,650	14.1±4.4	14.6±3.7	−0.43 (−1.75 to 0.89)
>65 to 80 yr	11,693	24.9±4.5	25.0±5.8	−0.08 (−2.92 to 2.76)
>80 yr	7,085	34.9±1.7	38.3±2.5	−3.41 (−6.25 to −0.56)
Case-mix group				
Medicine	17,798	31.4±4.8	32.7±3.9	−1.23 (−3.80 to 1.34)
Mental health	269	14.5±6.3	16.9±9.3	−3.48 (−12.9 to 5.96)
Pregnancy and childbirth	7,376	7.1±3.2	12.5±16.9	−0.50 (−2.13 to 1.13)
Surgery	18,183	14.9±3.7	15.2±3.9	−0.21 (−1.58 to 1.17)
Elixhauser quartile				
1	2,584	14.7±5.6	15.1±5.4	−0.18 (−3.23 to 2.87)
2	25,153	13.3±2.8	13.8±2.9	−0.40 (−1.21 to 0.40)
3	4,309	27.3±2.9	27.3±3.1	0.35 (−2.77 to 3.47)
4	11,580	36.0±4.4	38.2±5.0	−2.43 (−4.89 to 0.04)
ICU within 1 day				
No	39,254	19.6±3.7	20.4±3.3	−0.76 (−1.84 to 0.33)
Yes	4,372	31.6±6.8	35.1±7.0	−3.45 (−7.80 to 0.90)
Type of surgery				
Emergency	627	7.1±10.1	8.3±3.6	−1.22 (−9.59 to 7.15)
Urgent	1,008	13.0±2.1	11.4±3.8	1.76 (−2.46 to 5.98)
Elective	11,484	12.3±4.4	12.3±4.6	0.14 (−1.61 to 1.88)

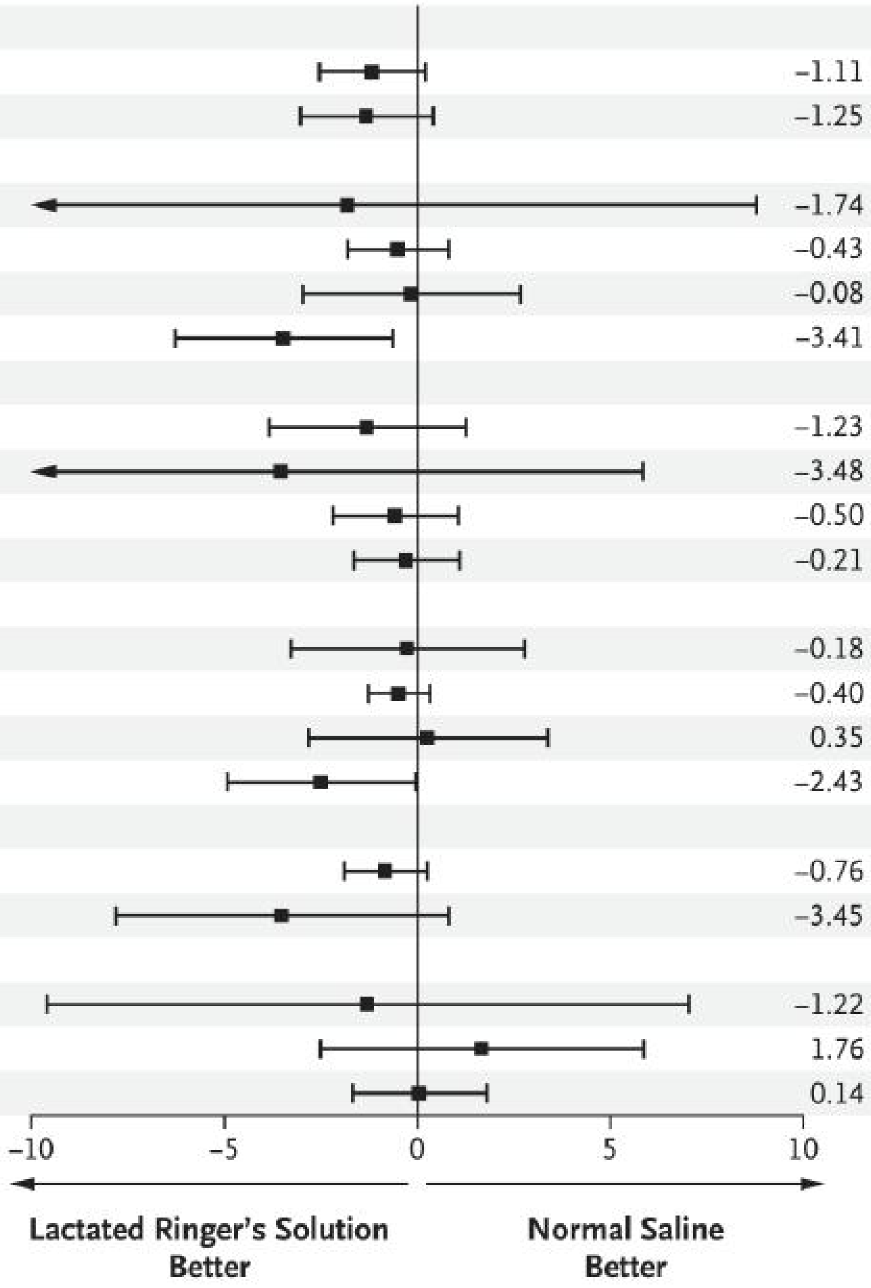


Figure 3. Subgroup Analyses.

Forest plots present results of the prespecified subgroup analyses of the primary outcome, a composite of death or hospital readmission within 90 days after the index hospital admission. Estimates are expressed as the mean (±SD) incidence of primary-outcome events across hospitals. Analyses were conducted with the use of stratified linear regression at the hospital level with trial period, assigned trial fluid, and hospital as independent variables. Case-mix groups were defined according to Canadian Institute for Health Information grouping methods on the basis of diagnosis, intervention, length of stay, and resource use. Elixhauser quartiles are based on scores on the Elixhauser Comorbidity Index, with higher scores indicating a higher number of coexisting conditions. ICU within 1 day denotes ad-

Figure 3. Subgroup forest plots of absolute risk difference of the primary outcome.

Subgroup Analyses Appraisal

- Subgroup Homogeneity: Results were generally consistent across all prespecified subgroups, including age, sex, comorbidity quartiles, and case-mix.
- Sepsis Subgroup: Sepsis patients (n=437) showed consistent results (Risk Difference -1.23%; 95% CI, -3.80% to 1.34%), with no statistically significant benefit.
- Traumatic Brain Injury: TBI patients (n=90) had a point estimate of -3.48% (95% CI, -12.9% to 5.96%). While small, it did not show the increased mortality concern seen in prior ICU trials (e.g., SAFE trial).
- Subgroup Discipline: Subgroups were prespecified but underpowered due to small subgroup sample sizes. All subgroup findings are strictly hypothesis-generating.

Results Critical Appraisal

EFFICACY STRENGTHS

- Large Database: Analyzing 43,626 patients through a centralized database ensures no loss to follow-up for the primary outcome.
- Clinical Consistency: Individual components (death, readmission, dialysis) were directionally consistent with the primary composite point estimate.

EFFICACY LIMITATIONS

- Underpowered: Stopping at 7 of 16 hospitals reduced the sample size, leaving the study underpowered to detect a 0.5% to 1.0% absolute risk reduction.
- Exposure Dilution: 21.8% non-compliance in the LR arm likely diluted the true physiological difference between balanced crystalloids and saline.
- Lack of Fluid Volume Data: The volume of fluid administered per patient was not reported, limiting our ability to assess dose-response relationships.

Results Verdict: The neutral primary composite outcome is definitive for this trial, but cannot rule out a small benefit due to the wide confidence intervals.

Risk of Bias 2 (RoB 2) Summary

Domain 1: Randomisation process Low Risk (Central block randomization stratified by hospital cluster; baseline covariates exceptionally well-balanced)	LOW RISK
Domain 2: Deviations from intended interventions Some Concerns (Open-label study; substantial non-adherence in the LR arm [78.2% adherence] and potential for cross-contamination)	SOME CONCERNS
Domain 3: Missing outcome data Low Risk (Registry-based tracking with zero loss to follow-up and complete primary/secondary outcomes)	LOW RISK
Domain 4: Measurement of the outcome Low Risk (Objective composite outcome of death/readmission validated with >99% accuracy in population databases)	LOW RISK
Domain 5: Selection of the reported result Low Risk (Followed a prepublished protocol and statistical analysis plan; registered prospectively)	LOW RISK
OVERALL RISK OF BIAS JUDGEMENT	**SOME CONCERNS**

GRADE Certainty of Evidence

GRADE FINAL CERTAINTY

MODERATE CERTAINTY

We are moderately confident in the effect estimate. The true effect is likely close to the estimate, but there is a possibility that it is substantially different.

Outcome: Composite of death or readmission within 90 days after index admission

Risk of Bias: NONE

No downgrade. Although Domain 2 had some concerns due to non-adherence, it represents real-world pragmatic implementation.

Inconsistency: NONE

No downgrade. Subgroup point estimates were directionally consistent across all cohorts.

Indirectness: NONE

No downgrade. Directly evaluated patient-important clinical outcomes (death, readmission, dialysis) in hospitalized patients.

Imprecision: DOWNGRADE 1 LEVEL

Downgraded due to wide confidence intervals that cross both the null and the clinically important threshold (1% absolute difference), caused by early study termination.

DISCUSSION

Key Trials, Guidelines, and Bayesian Reanalysis

Chronology of Key Crystalloid Trials

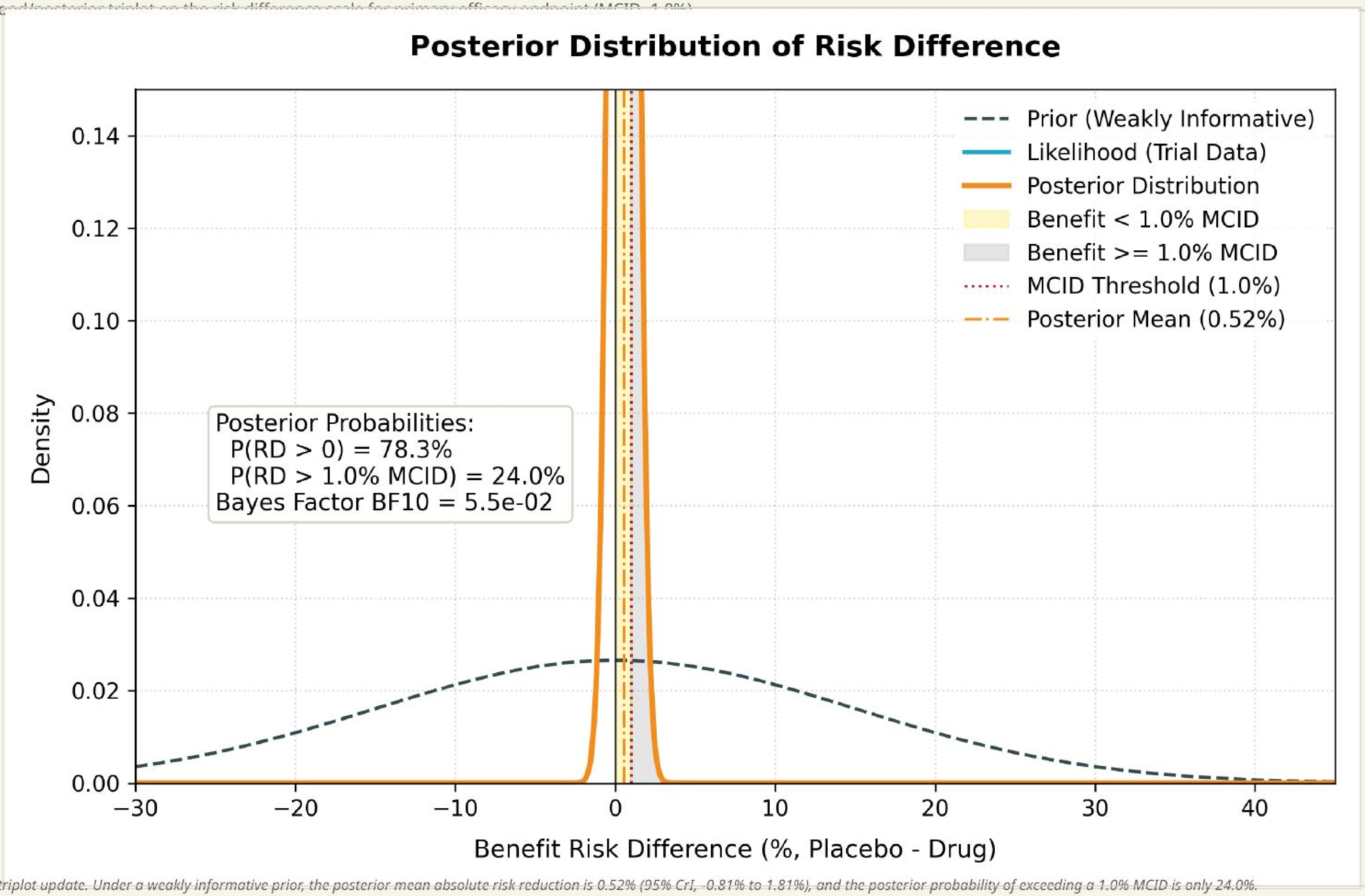
Study (Year)	Setting	Fluids Compared	Primary Outcomes & Key Findings
SAFE (2004)	ICU (n=6,997)	Albumin 4% vs. Normal Saline	No difference in 28-day mortality. Subgroup: increased mortality in Traumatic Brain Injury with albumin.
SPLIT (2015)	ICU (n=2,278)	Plasma-Lyte 148 vs. Normal Saline	No difference in AKI or renal replacement therapy. First large ICU trial comparing balanced crystalloids with saline.
SMART (2018)	ICU (n=15,802)	Balanced Crystalloids vs. Saline	Significant 1.1% absolute reduction in Major Adverse Kidney Events at 30 days (MAKE30) with balanced crystalloids.
SALT-ED (2018)	ED (n=13,347)	Balanced Crystalloids vs. Saline	Significant reduction in MAKE30 (4.7% vs. 5.6%) with balanced crystalloids in non-critically ill patients.
BASICS (2021)	ICU (n=10,520)	Plasma-Lyte 148 vs. Saline	No significant difference in 90-day mortality, showing equivalence of fluids in a pragmatic ICU setting.
FLUID (2025)	Hospital-wide (n=43,626)	Lactated Ringer's vs. Saline	No significant difference in 90-day death/readmission (-0.53% difference). Diluted by non-adherence; stopped early.

Current Guideline Consensus on Crystalloids

- Surviving Sepsis Campaign (2021): Recommends balanced crystalloids (LR or Plasma-Lyte) over normal saline for the resuscitation of adults with sepsis or septic shock (Weak recommendation, moderate-quality evidence).
- Clinical Practice Guidelines: Suggest balanced crystalloids in patients with diabetic ketoacidosis (DKA) or large-volume resuscitation to avoid hyperchloremic acidosis.
- General Medical Wards: No strong guideline recommendation exists for routine ward fluid choices. Individual patient factors (cardiac, renal, electrolyte status) dictate fluid choice.
- Consensus Verdict: Balanced crystalloids are preferred in critical care and sepsis, but normal saline remains acceptable and widely used in general medical wards.

Posterior Distribution of Risk Difference (Weakly Informative Prior)

Conjugate prior/likelih



Conjugate normal-normal triplot update. Under a weakly informative prior, the posterior mean absolute risk reduction is 0.52% (95% CrI, -0.81% to 1.81%), and the posterior probability of exceeding a 1.0% MCID is only 24.0%.

Posterior Efficacy under Alternative Priors

Sensitivity analysis of posterior probabilities under different prior assumptions

Prior Profile	Prior Mean (SD) *	Posterior Mean RD (95% CrI)	P(RD > 0)	P(RD > 1.0% MCID)
Weakly Informative (Default)	0.00 (2.0000)	0.52% (-0.81% to 1.81%)	78.3%	24.0%
Skeptical (MCID-derived)	0.00 (0.0472)	0.30% (-0.71% to 1.30%)	72.6%	8.5%
Enthusiastic (MCID-derived)	-0.06 (0.1154)	0.58% (-0.67% to 1.79%)	82.1%	25.3%
Historical (Phase 2 Data)	0.80 (0.4000)	0.39% (-0.95% to 1.68%)	72.0%	17.9%

** Note: Prior mean and SD are specified on the log-odds-ratio scale. Risk differences (RD) are placebo minus drug (positive values favor treatment). MCID is 1.0% absolute risk difference.*

APPLICABILITY & TAKEAWAYS

Local Settings, Ward Guidelines, and take-aways

Local Applicability: North Indian Patients

OPPORTUNITIES & EFFICACY

- Fluid Availability: Both Lactated Ringer's and normal saline are widely available and extremely cheap in public and private hospitals in Ghaziabad.
- Sepsis and Critical Illness: Sepsis is highly prevalent; using LR as the primary fluid aligns with global recommendations for resuscitation.
- Renal Safety: Avoiding hyperchloremic metabolic acidosis is particularly useful in patients with pre-existing renal impairment, which is common in our clinics.

BARRIERS & SAFETY CONCERNS

- Acidosis Misconceptions: LR contains lactate (not lactic acid), which is metabolized to bicarbonate. Clinicians sometimes mistakenly avoid LR in lactic acidosis.
- Lactate Monitoring: Using LR can transiently increase blood lactate levels, complicating monitoring in septic shock or hepatic failure patients.

Ghaziabad Clinical Context: While LR is generally safe and cheap, it cannot be assumed superior to saline for routine ward patients. Fluid choices should be individualized.

Clinical Takeaways for OPD, Wards, & ICU/HDU

OUTPATIENT CLINICS (OPD)

- Assess fluid and electrolyte status before prescribing home hydration or scheduling outpatient IV fluids.
- Individualize fluid choices: Normal saline is appropriate for volume depletion with hypochloremic alkalosis (e.g., severe vomiting).

INPATIENT WARDS

- Use Lactated Ringer's or balanced crystalloids as the default maintenance fluid to prevent hyperchloremic acidosis.
- Avoid routine large-volume saline unless treating severe hyponatremia or hypochloremia.
- Monitor serum electrolytes (sodium, chloride, potassium) every 24-48 hours during active fluid therapy.

ICU & HIGH DEPENDENCY (HDU)

- Prefer balanced crystalloids (LR or Plasma-Lyte) for initial resuscitation in sepsis or septic shock.
- Avoid LR in patients with severe hepatic failure (impaired lactate metabolism) or severe hyperkalemia (>5.5 mmol/L).
- Individualize therapy based on MAP, cardiac output, and real-world clinical endpoints.

Journal Club Discussion Questions

PRAGMATIC COMPLIANCE

1. How do we address fluid non-adherence and automatic substitution overrides in our clinical settings?

WARD VS ICU CHOICES

2. Should we standardise fluid policies hospital-wide, or should we separate general ward choices from ICU resuscitation choices?

LACTATE CONFUSION

3. What educational strategies can we use to resolve the clinical confusion surrounding the use of Lactated Ringer's in lactic acidosis?

POWER & TRIAL DESIGN

4. How does the early termination of the FLUID trial affect our trust in its neutral conclusions?

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